

# Evaluation of Feature Extraction Techniques in Content-Based Image Retrieval for Place recognition task

Linas Li

University of Twente  
Enschede, the Netherlands  
l.li-5@student.utwente.nl

## ABSTRACT

This study investigation probes the efficacy of Visual Place Recognition (VPR) within the domain of Content-Based Image Retrieval (CBIR). The study carefully examined three distinct feature extraction paradigms—Scale-Invariant Feature Transform (SIFT), Accelerated KAZE (AKAZE), and Oriented FAST and Rotated BRIEF (ORB), each coupled with a Bag-of-Visual-Words (BoVW) schema for vector representation. Concurrently, it compared a pre-trained Convolutional Neural Network (CNN), the ResNet50, as a modern feature extraction archetype. Employing the Mapillary Street-Level Sequences dataset, the research simulates diverse environmental conditions to test the resilience of VPR systems. The outcomes delineate the respective proficiencies and constraints of these methodologies, shedding light on their robustness, precision, and operational expediency. Notably, while SIFT remains a stalwart for dependability, CNN-based extractors excel in generalization, adeptly navigating the intricacies of visual data. This discourse advances the field of CBIR, charting a path towards refined place recognition capabilities.

## 1 INTRODUCTION

Information Retrieval (IR) systems are essential for searching relevant data within extensive collections of unstructured or semi-structured information [14]. With the expansion of these databases to scales that defy traditional cataloging efficacy, IR models become indispensable. Thus the information retrieval system ensures users to find relevant information from an organized collection of documents, which is designed to retrieve information about the existence or non-existence of documents relevant to a user query [24]. These systems have seen extensive application across domains such as web search [10], text retrieval [4], image retrieval [4] [10], and so on. Image retrieval, in particular, involves various methodologies to mine pictures from sizable digital image repositories.

Conventionally, image retrieval has hinged on utilizing metadata—captions, descriptions, titles, and keywords—to facilitate text-based searches [17]. However, metadata reliance necessitates manual image annotation, which is not only costly and labor-intensive but also subject to the inconsistency of human interpretation. The burgeoning daily proliferation of digital images from smartphones

and surveillance equipment has rendered such methods increasingly insufficient [12].

An alternative that has garnered significant research interest since the 1990s is Content-Based Image Retrieval (CBIR) [20]. This technique is designed to autonomously sift through an image database to identify images with visual content akin to a given query image [3]. Despite the high degree of automation, CBIR systems are often challenged by the 'semantic gap'—a term that encapsulates the disconnect between the low-level features (such as texture, color, and shape) used to describe images and the higher-level conceptual interpretations that users seek within these images [5]. This discrepancy can result in the retrieval of images that, while similar in visual features, are incongruent with the semantic intent of the search, thereby undermining the relevance of the results obtained.

The employment of machine learning has been extensively investigated as a promising approach to mitigate the 'semantic gap' in CBIR systems [28]. Deep learning's capability to assimilate complex patterns from expansive datasets has catalyzed the development of sophisticated neural architectures, such as VGG16 [23] and ResNet 50 [8], which have significantly advanced the field. These deep learning models are adept at abstracting high-level features from raw image data, offering a more nuanced alignment with human perceptual interpretations and thereby enhancing the relevance of image retrieval outcomes.

This research compares three distinct feature extraction algorithms: scale-invariant feature transform (SIFT) [16], accelerated KAZE (AKAZE) [1] [2] and oriented FAST and rotated BRIEF (ORB) [21], using a Bag-of-Visual-Words (BoW) model for vector encoding. Additionally, we examine the use of a pre-trained Convolutional Neural Network (CNN), the VGG-16, as a feature extractor, assessing its performance against the BoVW-encoded vectors.

Meanwhile, the specific feature representation—the encoding of an image into a vector descriptive of its contents—and the similarity metric used to gauge image resemblance are pivotal to retrieval efficacy. Our study aims to evaluate different configurations of these factors in the context of Visual Place Recognition (VPR), a process of matching images of a location under varying conditions. It's crucial to recognize that no single similarity metric universally fits all scenarios, hence the importance of testing a varied distance matrix to explore the most apt for the given application [27].

The structure of this paper is as follows: Section 2 reviews pertinent literature, Section 3 describes the dataset used, and Section 4 details the methodology, concentrating on the selected feature extractors and experimental framework. The principal findings are outlined in Section 5, where further analyzed and concluding remarks are presented in Section 6.

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## 2 RELATED WORK

The field of place recognition has witnessed a proliferation of methodologies tailored to diverse environments. Efforts in scene recognition have increasingly leveraged color [9] and texture [25] as foundational descriptors. To enhance recognition capabilities, traditional image descriptors, such as the Bag-of-Visual-Words, have been adapted across various domains [18]. Algorithms rooted in local features, notably SIFT [16], represent images through distinct elemental descriptors. These features are commonly aligned using methods like approximate nearest neighbor searches or through the employment of sophisticated data structures, for instance, the visual BoW framework as integrated in FAB-MAP 2.0 [7]. Despite exhibiting robustness to viewpoint shifts, these local feature-based approaches often falter in environments with significant alterations.

In a contemporary shift, holistic descriptors derived from Convolutional Neural Networks (CNNs) have emerged as potent tools for place recognition [6]. Sunderhauf et al. [26] introduced an approach that harnesses image descriptors from aggregated CNN outputs. This methodology, upon evaluation across various network layers, identified the initial convolutional layer as exceptionally resilient to changes in the scene, albeit with heightened sensitivity to alterations in viewpoint. Such descriptors have demonstrated formidable efficacy, particularly within challenging datasets such as the cross-seasonal Nordland.

## 3 DATASET

In this project, we utilize the Mapillary Street-Level Sequences (MSLS) dataset [29], a substantial corpus for long-term place recognition featuring 1.6 million street-level images from 30 diverse cities. Specifically, we focus on the subset of images capturing the cityscape of London, where 3291 images are used as map and 2692 images as queries. The accompanying 'london\_gt.h5' file encompasses exhaustive details on the images categorized as relevant and non-relevant, facilitating the evaluation process. This dataset poses formidable challenges due to its temporal and seasonal breadth, coupled with substantial variations in camera viewpoints, which together introduce a high degree of visual diversity and complexity to the task at hand. Fig. 1 presents a sample of the dataset employed in this project, showcasing the diversity and complexity of the images under analysis.



**Figure 1:** (a) Examples of image retrieval results: 1a an image accurately identified as relevant to the search query, 1b a relevant image retrieved from the map-based results, and 1c an example of a non-relevant image erroneously retrieved from map-based results, illustrating the challenges of accurate image retrieval in varying urban scenes.

## 4 METHODOLOGY

This section will provide a comprehensive discussion of various descriptors, along with an explanation of the experimental setups used in the study.

### 4.1 Feature extractors

Three local feature extraction algorithms have been employed to isolate keypoints within the images, alongside ResNet50 for global feature extraction.

**SIFT** Scale-Invariant Feature Transform (SIFT) selects salient points within an image through the detection of pronounced changes in brightness. Employing a multi-stage filtering approach, SIFT operates within a scale-space framework, utilizing a Difference of Gaussian-based keypoint detector in conjunction with a gradient orientation distribution descriptor [16]. This process yields robust 128-dimensional descriptors that maintain invariance to translations, rotations, and scaling, as well as to changes in illumination and blur. Widely applied in image matching, object detection, and 3D reconstruction, SIFT has demonstrated commendable performance in various scenarios [30]. Nonetheless, its computational intensity limits its applicability in real-time contexts.

**AKAZE** The Accelerated KAZE (AKAZE) algorithm functions analogously to its progenitor KAZE but introduces the Fast Explicit Diffusion (FED) for the expeditious generation of a non-linear scale-space [1] [2]. FED, predicated on a series of box filter decompositions, iteratively refines the image, starting with the original and subsequently utilizing the output from each cycle. AKAZE's efficiency is further augmented by the adoption of a binary descriptor, derived from an adapted Local Difference Binary (LDB) algorithm. This modified LDB approach extends BRIEF's methodology by executing binary tests around the keypoint's vicinity.

**ORB** Oriented FAST and Rotated BRIEF (ORB) is a binary descriptor, noted for its speed and efficiency, which builds upon the BRIEF descriptor with enhancements for rotation invariance and noise resistance [21]. It leverages the FAST feature detector, which identifies critical points in the image by examining the brightness variations around a pixel. The significance of FAST-detected keypoints is further assessed using the Harris corner measure. ORB conducts a pyramidal analysis, offering a multi-resolution inspection through different scaled versions of the image. It ascertains the orientation of each keypoint by analyzing intensity fluctuations, ultimately prioritizing the detection of the most pertinent features [22].

**ResNet50** Leyva-Vallina et al. [13] introduced a model that utilizes Generalized Contrastive Loss (GCL), a novel approach that capitalizes on graded similarity labels for training Siamese networks. This method circumvents the exhaustive process of pairwise mining by adjusting latent space distances to reflect the gradational similarities between images. Such optimization facilitates a more effective training regimen. Empirical results indicate that the fusion of GCL with graded similarity labels significantly enhances Visual Place Recognition (VPR), achieving results that are on par with or superior to those of conventional methods reliant on intensive pair mining and re-ranking strategies. The efficiency of this approach in handling data is particularly notable, presenting clear benefits for VPR tasks.

## 4.2 Setups

In the evaluation of each feature descriptor, various configurations were methodically examined. For SIFT, the experiment involved adjusting the quantity of prime features retained—specifically 50, 100, or 200—selected based on the strength of their local contrast. AKAZE featured distinct experimental conditions, utilizing threshold levels ( $th = 0.001, 0.002$ ) to modulate the count of detected keypoints; a higher threshold naturally resulted in fewer features being identified, and conversely, a lower threshold increased feature detection. For ORB, different counts of keypoints (50, 100, 200) were considered, prioritizing the top  $n$  keypoints based on the Harris corner response, unless fewer than  $n$  keypoints were detected, in which case all detected keypoints were utilized.

Within the framework of this research, a standard Bag of Words (BoW) model has been implemented for local feature extraction. This process commenced with the identification of keypoints and the extraction of a series of visual descriptors for each image in the database. Subsequently, K-means clustering was employed on the entire set of descriptors to categorize them into groups based on similarity, forming a visual vocabulary. The optimal number of clusters was determined to be 32, with the clustering process being repeated five times using different centroid seeds. The iteration yielding the lowest inertia was selected as the final model.

Each image in the dataset was then represented as a 32-dimensional vector, reflecting the histogram of visual word occurrences within that image. For each descriptor in the image, the nearest centroid was identified using the Euclidean distance metric, incrementing the corresponding index in the histogram by one. As a result, the aggregate of the vector elements equaled the total number of descriptors for that image. These BoWV vectors were subsequently normalized through z-score normalization to standardize the data.

The same normalization procedure was applied to the query images, which were standardized using the mean and variance parameters derived from the database images. This approach ensures that both the database and query images are represented in a consistent manner, facilitating more reliable image matching and retrieval.

Concurrently, the convolutional neural network (CNN) architecture adopted in this study utilizes ResNet50 as its backbone, integrating the Generalized Mean (GeM) pooling method [19]. The network’s optimization is facilitated by the AdamW optimizer [15]. Adaptations have been made to the terminal layers of the network; specifically, the last two layers were retrained using the MSLS dataset. This retraining was necessary because the previous iteration of the model was tailored to the ImageNet classifier, which categorizes images into 1,000 distinct classes [11]. The retraining aligns the model more closely with the requirements of the MSLS dataset and its unique classification challenges.

For each query image, the similarity assessment involved calculating the distance between its feature vector and those of the database images. Specifically, this study experimented with five distinct distance metrics to evaluate their effectiveness in feature vector comparison:

**Euclidean distance** gauges the direct line distance in Euclidean space between two vectors, calculated via the root of the sum of squared discrepancies between corresponding vector

components.

$$d_{\text{euclidean}}(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (1)$$

**Cosine distance** measures vector orientation disparity, the cosine distance ascertains the cosine value of the angle between two vectors, independent of their magnitude.

$$d_{\text{cosine}}(x, y) = 1 - \frac{\sum_{i=1}^n x_i y_i}{\sqrt{\sum_{i=1}^n x_i^2} \cdot \sqrt{\sum_{i=1}^n y_i^2}} \quad (2)$$

**Minkowski distance** is especially sensitive to data variations, accentuating more pronounced differences through the cubic power.

$$d_{\text{minkowski}}(x, y) = \left( \sum_{i=1}^n |x_i - y_i|^3 \right)^{1/3} \quad (3)$$

**Manhattan distance** tallies the sum of absolute differences along Cartesian coordinates, mimicking a city grid, due to its application in grid-like pathfinding.

$$d_{\text{manhattan}}(x, y) = \sum_{i=1}^n |x_i - y_i| \quad (4)$$

**Hamming distance** is used for binary data and enumerates the distinct positions between two strings, effectively the count of bit alterations needed to transmute one into the other.

$$d_{\text{hamming}}(x, y) = \frac{1}{n} \sum_{i=1}^n [x_i \neq y_i] \quad (5)$$

Subsequent to computing the distance metrics, the images within the database were sequenced in ascending order, reflecting their proximity—from nearest to most distant—to the feature vector of the query image. The performance evaluation of the retrieval systems was predicated upon two key metrics:

**Top recall at k** computes the percentage of queries for which the system successfully retrieves at least one pertinent image from the map within the top K ranked images. R@K was specifically calculated for  $K = 1, 5, 10$ .

**Mean average precision** is the aggregate of the Average Precision (AP) for each query, averaged over the total number of queries. Each query’s AP is an aggregate measure that integrates the precision-recall curve into a singular value, reflecting the mean of all precision levels achieved at each threshold. Thus, the mAP serves as an estimation of the mean area under the precision-recall curve across all queries.

## 5 RESULTS

Table 1 elucidates the relative efficacy of diverse distance metrics across a spectrum of feature descriptors on the MSLS London Dataset. Each descriptor’s performance is rigorously quantified via Top Recall at K (R@K) for  $K=1,5,10$  and Mean Average Precision (mAP). The most precise results are accentuated in bold, indicating superior precision among the configurations tested. Furthermore, the values underlined signify the most favorable outcomes within each distance metric category for the respective feature descriptor.

The assessment incorporates a breadth of metrics, including Euclidean, Cosine, Minkowski, Manhattan, and Hamming distances. Feature descriptors such as SIFT were evaluated with variable features (50, 100, 200); AKAZE with distinct thresholding levels (0.001, 0.002); and ORB with a set number of keypoints (50, 100, 200). For comparative analysis, the performance of a CNN employing ResNet50 with GeM pooling GCL model was also appraised.

Relative to the local feature extractors, the CNN model exhibited a notable enhancement in retrieval accuracy, achieving an absolute increase in R@1, R@5, and R@10 of 0.0484, 0.114, and 0.1158, respectively, when juxtaposed with the top-performing scores. Moreover, the mean Average Precision (mAP) of the CNN model demonstrated a substantial rise of 0.7299 over its local feature descriptor counterparts.

## 6 DISCUSSION AND CONCLUSION

The experimental results reveal that for SIFT, AKAZE, and ORB descriptors, an increase in the number of detected keypoints correlates with improved retrieval performance. This improvement is attributed to the richer representation of the image content provided by a greater number of significant local features, which in turn facilitates the retrieval of more relevant examples in response to a query. An exception to this trend is observed when employing the Hamming distance due to its binary nature. Notably, SIFT outperforms other feature extractors, achieving the highest scores.

The Mean Average Precision (mAP) exhibits the least variability across the evaluated algorithms. For local feature extractors, mAP spans from 0.0229 to 0.0401, with the CNN model reaching up to 0.0770. This lower variance in mAP is explained by its sensitivity to the precise ranking of relevant images, unlike Recall at K (R@K), which only considers the presence of relevant items within the top K results regardless of their exact positions. While R@K is averaged across queries, mAP represents an average of precision at successive ranks, making it inherently more stable and an indicator of the precise retrieval order. Therefore, the choice between R@K and mAP as a performance metric should be guided by the specific needs of the application—R@K is preferred when the presence of relevant items among the top results suffices, while mAP is advantageous when the exact order of these items is of essence.

This study conducts an evaluation of three established feature extraction algorithms and a contemporary CNN model in the context of Visual Place Recognition (VPR), utilizing a selection of the MSLS dataset that encompasses imagery from London. Within the realm of traditional computer vision techniques, we tailored a particular parameter to yield three distinct configurations. For SIFT and ORB, we adjusted the count of retained features, whereas for AKAZE, we altered the threshold of the descriptors. The database images were subsequently retrieved using a quartet of distance metrics: Euclidean, cosine, Minkowski, Manhattan and Hamming. The efficacy of the Image Retrieval task was assessed through two metrics prevalent in this domain: Top Recall@K and Mean Average Precision.

Future investigations could extend to the exploration of the retrieval performance variability when other parameters are modified.

Employing alternative clustering algorithms such as spectral clustering, hierarchical clustering or normalization techniques could provide valuable insights into their impact on system efficiency.

Furthermore, deep learning strategies could pivot towards 'learning to rank' methodologies or the adoption of deeper network backbones to scrutinize their performance relative to that achieved using generalized contrastive loss. Additionally, the integration of transformer architectures, which have recently achieved significant success in a myriad of computer vision applications, presents a compelling avenue for research. The increasing deployment of such architectures in real-world applications amplifies the importance of understanding their potential in enhancing.

Overall, the results substantiate the efficacy of deep learning methodologies in generating robust feature representations for Content-Based Image Retrieval (CBIR) tasks, offering a promising solution to the perennial issue of the semantic gap in image retrieval. Nonetheless, these methods are their limitation; they typically require a large dataset and considerable computational expensive. Conversely, classical computer vision techniques has shown a trend to be more light weighted and less relied on the training dataset.

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**Table 1:** Comparative Analysis of Method Performance Across Diverse Distance Metrics on the MSLS London Dataset with Optimal Results Highlighted in Bold and underlines

| Metric       | Similarity | SIFT          |               |               | AKAZE         |               | ORB           |               |               | ResNet50 GCL  |
|--------------|------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|              |            | 50            | 100           | 200           | 0.001         | 0.002         | 50            | 100           | 200           |               |
| TopRecall@1  | euclidean  | 0.0093        | 0.0178        | 0.0234        | 0.0152        | 0.0126        | 0.0074        | 0.0067        | 0.0074        | <b>0.0770</b> |
|              | cosine     | <u>0.0119</u> | <u>0.0200</u> | <b>0.0286</b> | <u>0.0208</u> | <u>0.0156</u> | <u>0.0089</u> | <u>0.0108</u> | <u>0.0093</u> | -             |
|              | minkowski  | 0.0067        | 0.0163        | <b>0.0197</b> | 0.0134        | 0.0126        | 0.0067        | 0.0071        | 0.0078        | -             |
|              | manhattan  | 0.0111        | 0.0186        | <b>0.2006</b> | 0.0126        | 0.0126        | 0.0082        | 0.0093        | 0.0082        | -             |
|              | hamming    | 0.0074        | <b>0.0082</b> | 0.0074        | 0.0026        | 0.0045        | 0.0022        | 0.0019        | 0.0045        | -             |
| TopRecall@5  | euclidean  | 0.0350        | 0.0498        | 0.0654        | 0.0446        | 0.0401        | 0.0227        | 0.0279        | 0.0316        | <b>0.1920</b> |
|              | cosine     | <u>0.0375</u> | <u>0.0579</u> | <b>0.0780</b> | <u>0.0468</u> | <u>0.0420</u> | 0.0227        | 0.0305        | <u>0.0353</u> | -             |
|              | minkowski  | 0.0323        | 0.0479        | <b>0.0654</b> | 0.0416        | 0.0350        | 0.0215        | 0.0249        | 0.0290        | -             |
|              | manhattan  | 0.0353        | 0.0509        | <b>0.0656</b> | 0.0409        | 0.0386        | <u>0.0271</u> | <u>0.0316</u> | 0.0319        | -             |
|              | hamming    | <b>0.0282</b> | 0.0267        | 0.0227        | 0.0189        | 0.0182        | 0.0182        | 0.0178        | 0.0171        | -             |
| TopRecall@10 | euclidean  | 0.0560        | 0.0873        | 0.0992        | 0.0672        | 0.0628        | 0.0397        | 0.0520        | 0.0565        | <b>0.2310</b> |
|              | cosine     | <u>0.0646</u> | <u>0.0902</u> | <b>0.1152</b> | <u>0.0780</u> | <u>0.0661</u> | 0.0409        | 0.0513        | <u>0.0617</u> | -             |
|              | minkowski  | 0.0565        | 0.0821        | <b>0.0925</b> | 0.0635        | 0.0620        | 0.0390        | 0.0453        | 0.0535        | -             |
|              | manhattan  | 0.0617        | 0.0862        | <b>0.1032</b> | 0.0600        | 0.0654        | <u>0.0427</u> | <u>0.0550</u> | 0.0598        | -             |
|              | hamming    | 0.0501        | <b>0.0516</b> | 0.0461        | 0.0342        | 0.0364        | 0.0390        | 0.0308        | 0.0286        | -             |
| mAP          | euclidean  | 0.0270        | 0.0330        | 0.0359        | 0.0309        | 0.0293        | 0.0249        | 0.0262        | 0.0268        | <b>0.0770</b> |
|              | cosine     | <u>0.0300</u> | <u>0.0358</u> | <b>0.0401</b> | <u>0.0336</u> | <u>0.0316</u> | <u>0.0262</u> | <u>0.0276</u> | <u>0.0284</u> | -             |
|              | minkowski  | 0.0270        | 0.0322        | <b>0.0347</b> | 0.0300        | 0.0286        | 0.0242        | 0.0258        | 0.0265        | -             |
|              | manhattan  | 0.0281        | 0.0331        | <b>0.0351</b> | 0.0300        | 0.0291        | 0.0256        | 0.0267        | 0.0270        | -             |
|              | hamming    | <b>0.0254</b> | 0.0252        | 0.0243        | 0.0233        | 0.0239        | 0.0232        | 0.0230        | 0.0229        | -             |

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